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1**Calcium homeostasis**

COMP

You should be able to: Trace the response to falling plasma calcium through parathyroid hormone, calcitriol, and the actions on bone, kidney, and intestine, and state the role of calcitonin.

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2**Kidney functions**

COMP

You should be able to: List the homeostatic functions of the kidney including water and electrolyte balance, acid base regulation, waste excretion, blood pressure control, and hormone production.

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3**Nephron structure and function**

COMP

You should be able to: Match each nephron segment to its dominant transport function and distinguish a cortical from a juxtamedullary nephron by capability.

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4**Three renal processes**

COMP

You should be able to: Define filtration, reabsorption, secretion, and excretion and write the equation that relates them for any solute.

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5**Filtration membrane and selectivity**

COMP

You should be able to: Describe the three layers of the filtration membrane and predict which plasma components appear in the filtrate based on size and charge.

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6**Net filtration pressure and GFR**

COMP

You should be able to: Calculate net filtration pressure from glomerular hydrostatic pressure, capsular pressure, and colloid osmotic pressure and predict the effect of each change on glomerular filtration rate.

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7**Regulation of GFR**

COMP

You should be able to: Compare myogenic autoregulation, tubuloglomerular feedback, sympathetic control, and hormonal control and predict the effect of afferent and efferent arteriolar constriction on filtration rate.

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8**Renal clearance**

COMP

You should be able to: Calculate the clearance of a substance and use inulin and creatinine clearance to estimate glomerular filtration rate.

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9**Clearance and handling inference**

COMP

You should be able to: Use a clearance value relative to glomerular filtration rate to determine whether a substance is net reabsorbed or net secreted.

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10**Proximal tubule reabsorption**

COMP

You should be able to: Explain how the sodium gradient drives reabsorption of glucose, amino acids, and water in the proximal tubule and state why reabsorption there is described as isosmotic.

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11**Transport maximum and renal threshold**

COMP

You should be able to: Distinguish transport maximum from renal threshold and explain the appearance of glucose in urine when plasma glucose is elevated.

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12**Tubular secretion**

COMP

You should be able to: State the purpose of tubular secretion and name substances secreted including hydrogen ion, potassium, and organic anions such as drugs.

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13**Countercurrent mechanism**

COMP

You should be able to: Explain how the countercurrent multiplier of the loop of Henle and the countercurrent exchanger of the vasa recta build and preserve the medullary osmotic gradient.

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14**Antidiuretic hormone and urine concentration**

COMP

You should be able to: Trace the response to dehydration from osmoreceptor firing through antidiuretic hormone release to aquaporin insertion and concentrated urine.

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15**Renin angiotensin aldosterone system**

COMP

You should be able to: Trace the renin angiotensin aldosterone pathway from the stimulus for renin release to the effects of angiotensin II and aldosterone on vessels, tubules, and blood pressure.

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16**Natriuretic peptides**

COMP

You should be able to: Explain how atrial natriuretic peptide opposes the renin angiotensin aldosterone system and state the stimulus for its release.

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17**Potassium handling**

COMP

You should be able to: Explain how aldosterone and plasma potassium regulate potassium secretion in the collecting duct and predict the effect of a potassium wasting diuretic.

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18**Micturition**

COMP

You should be able to: Trace the micturition reflex and explain how the internal and external sphincters allow voluntary control.

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19**Urinalysis**

COMP

You should be able to: Perform a urinalysis measuring specific gravity, pH, glucose, protein, ketones, and blood and relate any abnormal finding to the renal process that failed.

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20**Renal function calculation**

COMP

You should be able to: Calculate glomerular filtration rate, clearance, and filtered load from a data set and interpret the values in a clinical case.

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21**Volume and osmolarity disturbances**

COMP

You should be able to: Classify a disturbance as volume depletion or overload and as hypoosmotic, isosmotic, or hyperosmotic and predict the resulting fluid shift between compartments.

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Dr. Sharilyn Rennie